REG NO:2025115008

NAME: KEERTHIVARMAN.KV

DATA STRUCTURES HOMEWORK 1

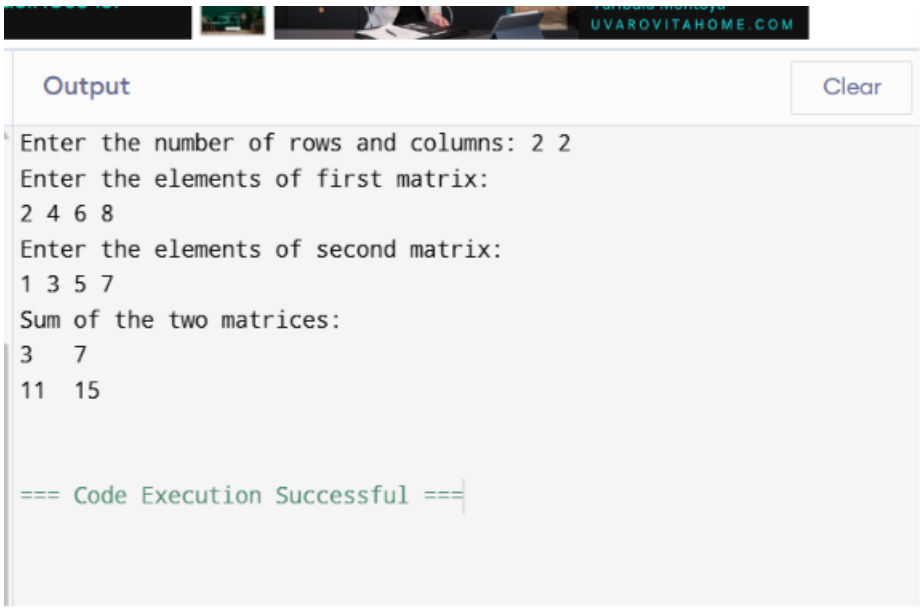
1. WRITE A C PROGRAM FOR SUM OF TWO MATRICES.

```
#include <stdio.h>
int main()
{
   int a[10][10],b[10][10],sum[10][10];
   int rows,cols;
   int i,j;

   printf("enter the number of rows and columns:");
   scanf("%d %d",&rows,&cols);

   printf("Enter the elements of first matrix:\n");
   for(i=0;i<rows;i++)
   {
      for(j=0;j<cols;j++)
      {
         scanf("%d",&a[i][j]);
      }
   }
   printf("Enter the elements of second matrix:\n");
   for(i=0;i<rows;i++)
   {
      for(j=0;j<cols;j++)
      {
         scanf("%d",&b[i][j]);
      }
   }
   for(i=0;i<rows;i++)
   {
      for(j=0;j<cols;j++)
      {
         sum[i][j]=a[i][j]+b[i][j];
      }
   }
```

```
    printf("Sum of the Matrices:\n");
    for(i=0;i<rows;i++)
    {
      for(j=0;j<cols;j++)
      {
        printf("%d\t",sum[i][j]);
      }
      printf("\n");
    }
    return 0;
}
```

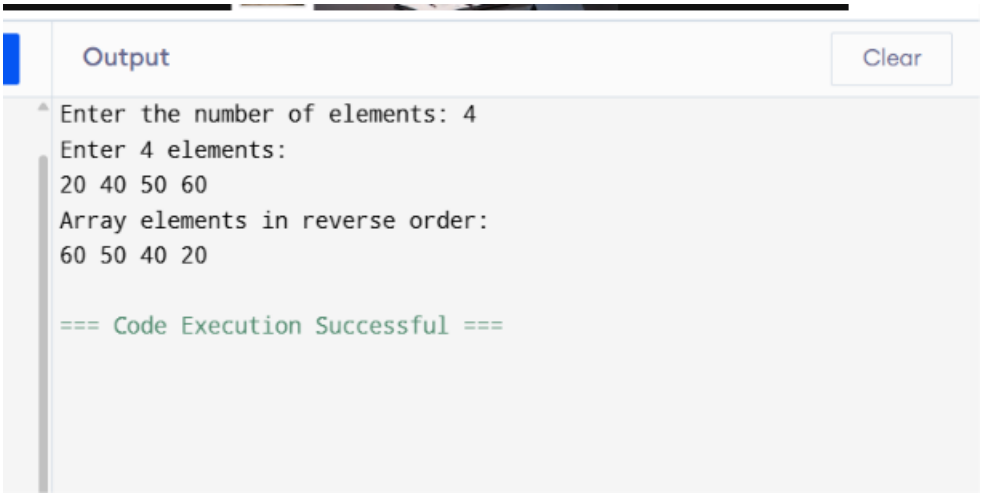


2. WRITE A C PROGRAM TO READ n NUMBER OF VALUES IN AN ARRAY & DISPLAY THEM IN REVERSE ORDER.

```
#include<stdio.h>
int main()
{
  int a[100];
  int n,i;
  printf("Enter the number of Elements:");
  scanf("%d",&n);

  printf("Enter %d Elements:\n",n);
  for(i=0;i<n;i++)
  {
    scanf("%d",&a[i]);
```

```
    }
    printf("Array elements in reverse order:\n");
    for(i=n-1;i>=0;i--)
    {
       printf("%d ",a[i]);
    }
    return 0;
}
```



3. WRITE A C PROGRAM TO COUNT THE TOTAL NUMBER OF DUPLICATE ELEMENTS IN AN ARRAY.

```
#include <stdio.h>

int main()
{
   int a[100];
   int n, i, j;
   int count = 0;

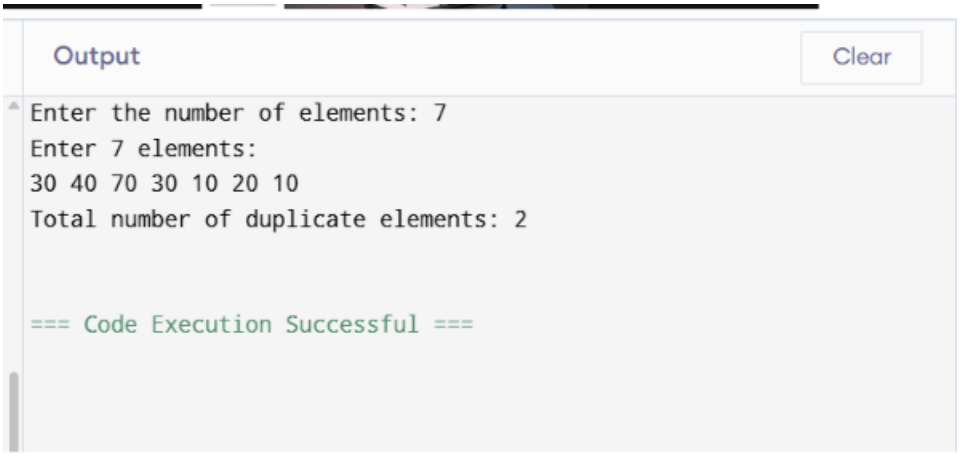
   // Read the number of elements
   printf("Enter the number of elements: ");
   scanf("%d", &n);

   // Read array elements
   printf("Enter %d elements:\n", n);
   for (i = 0; i < n; i++)
   {
      scanf("%d", &a[i]);
   }
```

```
    // Count duplicate elements
    for (i = 0; i < n; i++)
    {
        for (j = i + 1; j < n; j++)
        {
            if (a[i] == a[j])
            {
                count++;
                break;
            }
        }
    }

    // Display the result
    printf("Total number of duplicate elements: %d\n", count);

    return 0;
}
```

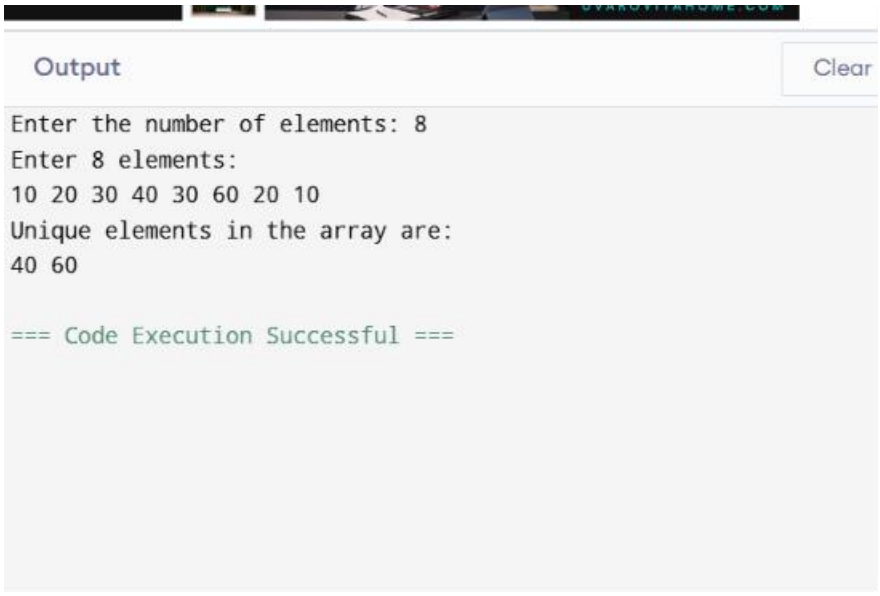


4. WRITE A C PROGRAM TO PRINT UNIQUE ELEMENT IN AN ARRAY.

```
#include<stdio.h>
int main()
{
    int a[100];
    int n,i,j,count;

    printf("Enter the number of elements:");
    scanf("%d",&n);
```

```
  printf("Enter %d elements:\n",n);
  for(i=0;i<n;i++)
  {
    scanf("%d",&a[i]);
  }
  printf("Unique elements in the array are :\n");
  for(i=0;i<n;i++)
  {
    count=0;
    for(j=0;j<n;j++)
    {
      if(a[i]==a[j])
      {
        count++;
      }
    }
    if(count==1)
    {
      printf("%d ",a[i]);
    }
  }
  return 0;
}
```



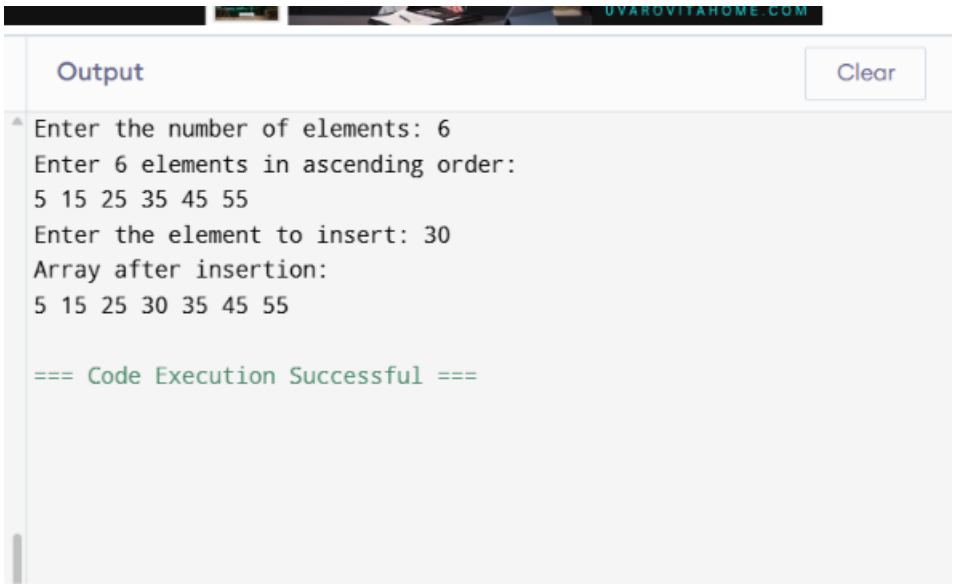
5. WRITE A C PROGRAM TO INSERT AN ELEMENT INTO A SORTED ARRAY.

```
#include<stdio.h>
int main()
```

```
{
   int a[100];
   int n,i,element,pos;

   printf("Enter the number of elements:");
   scanf("%d",&n);

   printf("Enter %d elements in ascending order:\n",n);
   for(i=0;i<n;i++)
   {
      scanf("%d",&a[i]);
   }
   printf("Enter the elements to insert:");
   scanf("%d",&element);
   pos=0;
   while(pos<n && a[pos]<element)
   {
      pos++;
   }
   for(i=n;i>pos;i--)
   {
      a[i]=a[i-1];
   }
   a[pos]=element;
   n++;
   printf("Array after insertion:\n");
   for(i=0;i<n;i++)
   {
      printf("%d ",a[i]);
   }
   return 0;
}
```



6. WRITE A C PROGRAM TO INSERT A ELEMENT IN UNSORTED ARRAY.

```
#include<stdio.h>
int main()
{
  int a[100];
  int n,i,element,position;

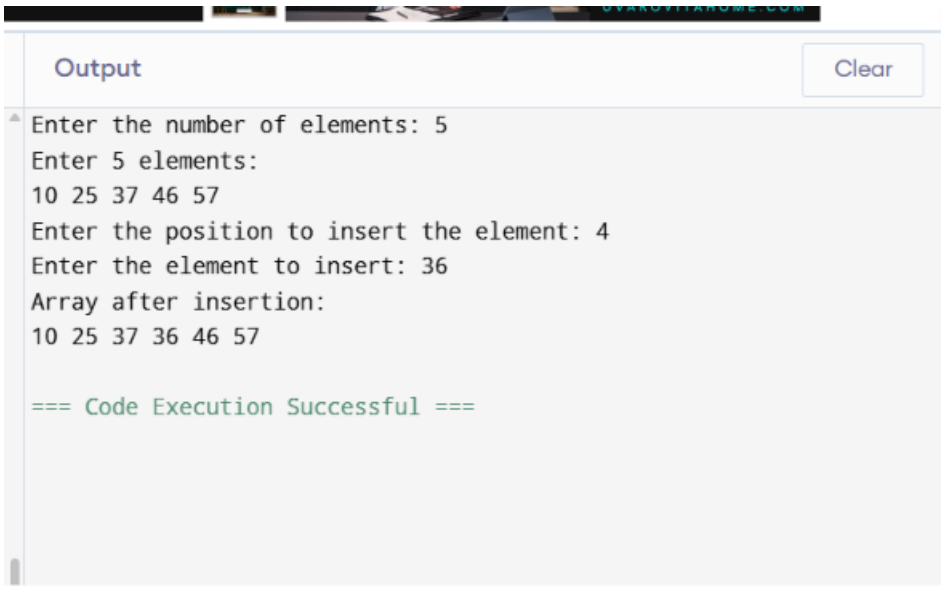
  printf("Enter the number of elements:");
  scanf("%d",&n);

  printf("Enter %d elements:\n",n);
  for(i=0;i<n;i++)
  {
    scanf("%d",&a[i]);
  }
  printf("Enter the position to insert the element:");
  scanf("%d",&position);

  printf("Enter the element to insert:");
  scanf("%d",&element);

  if(position<1 || position>n+1)
  {
    printf("Invalid Position.\n");
    return 0;
  }
  for(i=n;i>position;i--)
```

```
    {
      a[i]=a[i-1];
    }
    a[position-1]=element;
    n++;
    printf("Array after insertion:\n");
    for(i=0;i<n;i++)
    {
      printf("%d ",a[i]);
    }
    return 0;
}
```



7. WRITE A C PROGRAM FOR MATRIX MULTIPLICATION.

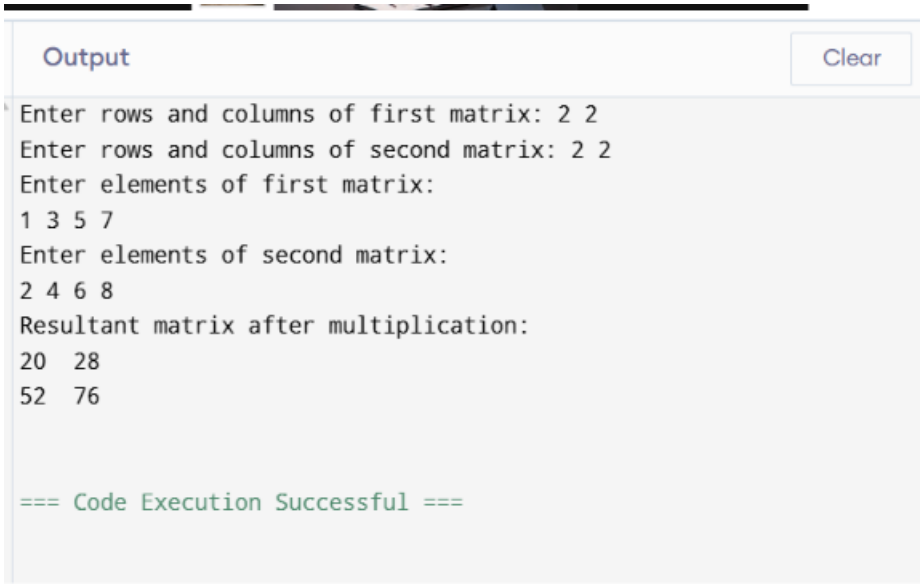
```
#include<stdio.h>
int main()
{
   int a[10][10],b[10][10],product[10][10];
   int r1,c1,r2,c2;
   int i,j,k;

   printf("Enter rows and columns of first matrix:");
   scanf("%d %d",&r1,&c1);

   printf("Enter rows and columns of second matrix:");
   scanf("%d %d",&r2,&c2);
```

```
if(c1!=r2)
{
   printf("Matrix Mulftiplication is not possible.\n");
   return 0;
}
printf("Enter elements of first matrix:\n");
for(i=0;i<r1;i++)
{
   for(j=0;j<c1;j++)
   {
      scanf("%d",&a[i][j]);
   }
}
printf("Enter elements of second matrix:\n");
for(i=0;i<r2;i++)
{
   for(j=0;j<c2;j++)
   {
      scanf("%d",&b[i][j]);
   }
}
for(i=0;i<r1;i++)
{
   for(j=0;j<c2;j++)
   {
      product[i][j]=0;
   }
}
for(i=0;i<r1;i++)
{
   for(j=0;j<c2;j++)
   {
      for(k=0;k<c1;k++)
      {
         product[i][j]+=a[i][k]*b[k][j];
      }
   }
}
printf("Resultant Matrix:\n");
for(i=0;i<r1;i++)
{
```

```
    for(j=0;j<c2;j++)
    {
      printf("%d\t",product[i][j]);
    }
    printf("\n");
  }
  return 0;
}
```

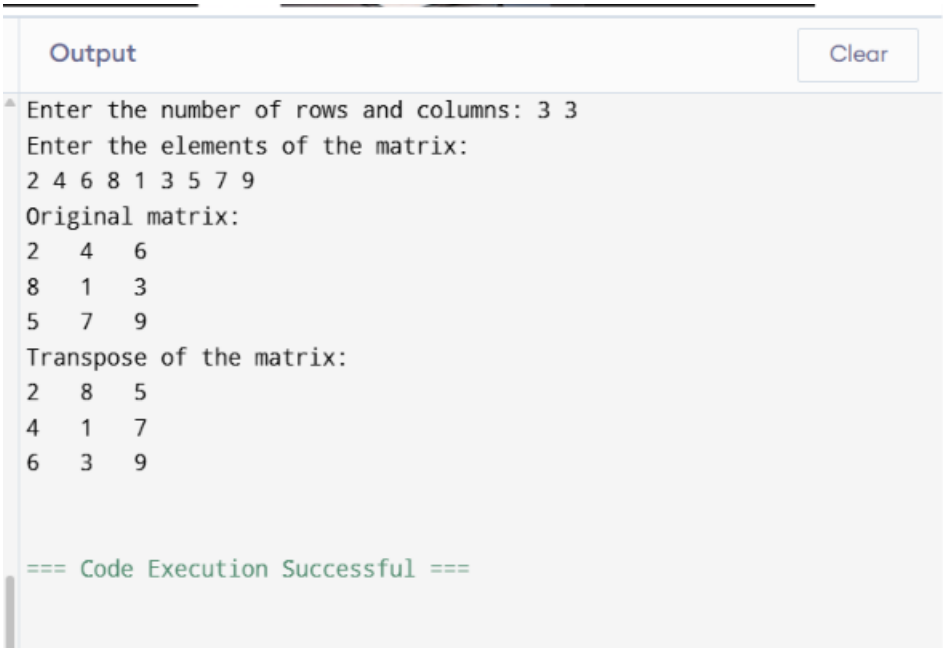


8. WRITE A C PROGRAM FOR MATRIX TRANSPOSE.

```
#include<stdio.h>
int main()
{
  int a[10][10],transpose[10][10];
  int rows,cols;
  int i,j;

  printf("Enter the number of rows and columns:");
  scanf("%d %d",&rows,&cols);
  printf("Enter the elements of the matrix:");
  for(i=0;i<rows;i++)
  {
    for(j=0;j<cols;j++)
    {
      scanf("%d",&a[i][j]);
    }
```

```
    }
    for(i=0;i<rows;i++)
    {
        for(j=0;j<cols;j++)
        {
            transpose[j][i]=a[i][j];
        }
    }
    printf("Original Matrix:\n");
    for(i=0;i<rows;i++)
    {
        for(j=0;j<cols;j++)
        {
            printf("%d\t",a[i][j]);
        }
        printf("\n");
    }
    printf("Transpose of the matrix:\n");
    for(i=0;i<cols;i++)
    {
        for(j=0;j<rows;j++)
        {
            printf("%d\t",transpose[i][j]);
        }
        printf("\n");
    }
    return 0;
}
```



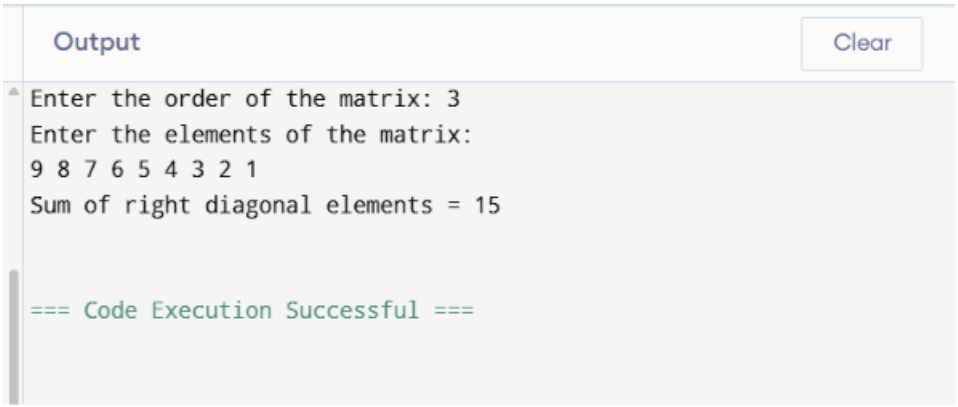
9. WRITE A C PROGRAM FOR RIGHT DIAGONAL SUM OF THE MATRIX.

```
#include<stdio.h>
int main()
{
  int a[10][10];
  int n,i,j,sum=0;

  printf("Enter the order of the matrix:");
  scanf("%d",&n);

  printf("Enter elements of matrix:");
  for(i=0;i<n;i++)
  {
    for(j=0;j<n;j++)
    {
      scanf("%d",&a[i][j]);
    }
  }
  for(i=0;i<n;i++)
  {
    sum=sum+a[i][n-1-i];
  }
  printf("Sum of right diagonal elements= %d\n",sum);
  return 0;
```

}



10. WRITE A C PROGRAM TO CHECK FOR IDENTITY MATRIX.

```
#include<stdio.h>
int main()
{
   int a[10][10];
   int n,i,j;
   int isIdentity=1;

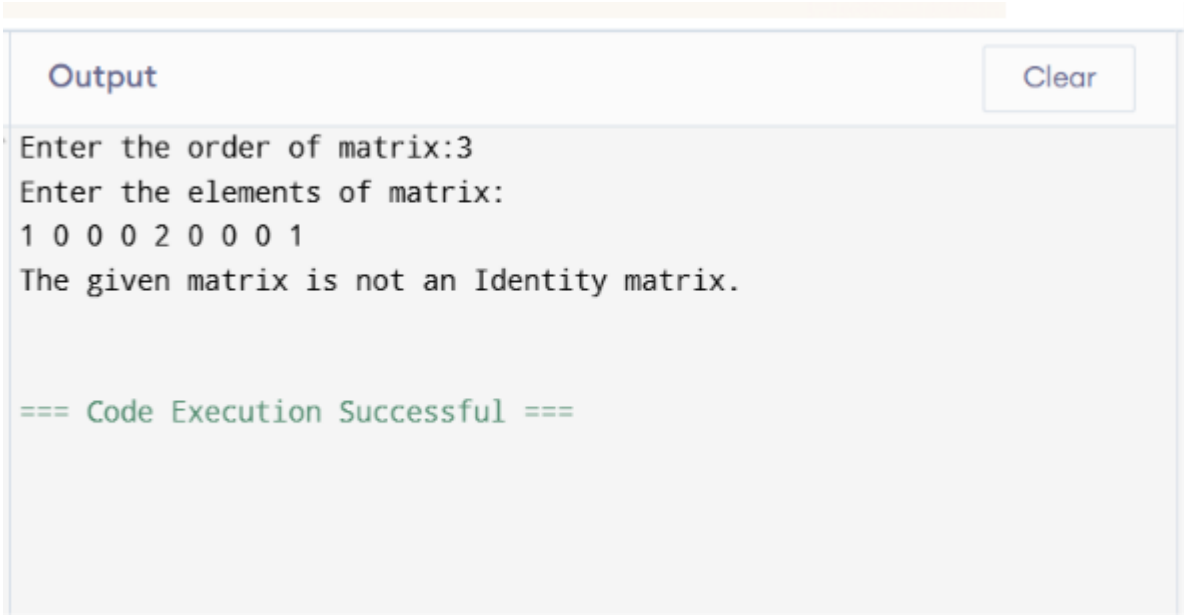
   printf("Enter the order of matrix:");
   scanf("%d",&n);
   printf("Enter the elements of matrix:\n");
   for(i=0;i<n;i++)
   {
      for(j=0;j<n;j++)
      {
         scanf("%d",&a[i][j]);
      }
   }
   for(i=0;i<n;i++)
   {
      for(j=0;j<n;j++)
      {
         if(i==j)
         {
            if(a[i][j]!=1)
            {
               isIdentity=0;
            }
         }
```

```
        else
        {
          if(a[i][j]!=0)
          {
            isIdentity=0;
          }
        }
      }
    }
    if(isIdentity==1)
    {
      printf("The given matrix is an Identity matrix.\n");
    }
    else
    {
      printf("The given matrix is not an Identity matrix.\n");
    }
    return 0;
}
```

Output Clear

```
Enter the order of the matrix: 3
Enter the elements of the matrix:
1 0 0
0 1 0
0 0 1
The given matrix is an identity matrix.


=== Code Execution Successful ===
```



11. WRITE A C PROGRAM TO FIND NUMBER OF VOWELS, CONSONANTS, DIGITS AND WHITE SPACES.

```
#include <stdio.h>

int main()
{
   char str[200];
   int i;
   int vowels = 0, consonants = 0, digits = 0, spaces = 0;

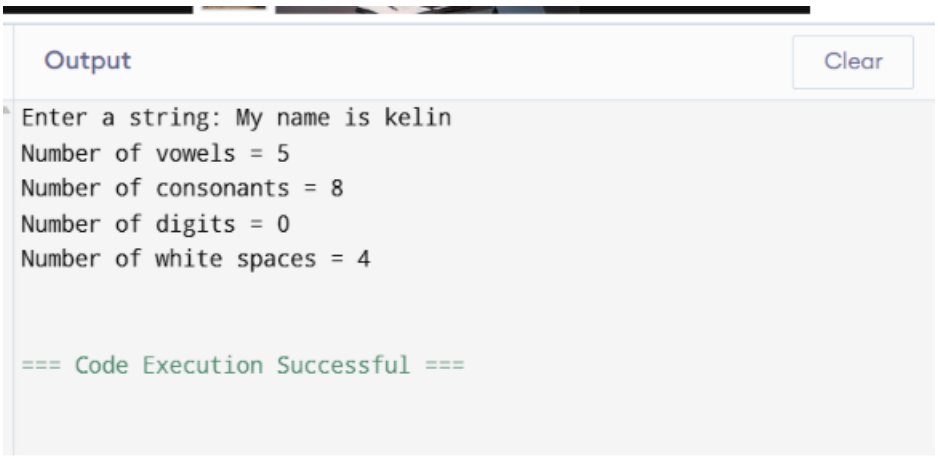
   printf("Enter a string:");
   fgets(str, sizeof(str), stdin);

   for (i = 0; str[i] != '\0'; i++)
   {
      if ((str[i] >= 'A' && str[i] <= 'Z') ||
         (str[i] >= 'a' && str[i] <= 'z'))
      {
         if (str[i] == 'A' || str[i] == 'E' || str[i] == 'I' ||
            str[i] == 'O' || str[i] == 'U' ||
            str[i] == 'a' || str[i] == 'e' || str[i] == 'i' ||
            str[i] == 'o' || str[i] == 'u')
         {
            vowels++;
         }
         else
```

```
        {
            consonants++;
        }
    }
    else if (str[i] >= '0' && str[i] <= '9')
    {
        digits++;
    }
    else if (str[i] == ' ' || str[i] == '\t' || str[i] == '\n')
    {
        spaces++;
    }
  }

  printf("Number of vowels = %d\n", vowels);
  printf("Number of consonants = %d\n", consonants);
  printf("Number of digits = %d\n", digits);
  printf("Number of white spaces = %d\n", spaces);

  return 0;
}
```



12. WRITE A C PROGRAM FOR FREQUENCY OF A CHARACTER IN STRING.

```
#include <stdio.h>

int main()
{
  char str[100];
  char ch;
  int i, frequency = 0;
```

```
    printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);

    printf("Enter the character to find: ");
    scanf(" %c", &ch);

    for (i = 0; str[i] != '\0'; i++)
    {
        if (str[i] == ch)
        {
            frequency++;
        }
    }

    printf("Frequency of '%c' = %d\n", ch, frequency);

    return 0;
}
```



13. WRITE A C PROGRAM TO COUNT A WORDS IN A STRING.

```
#include <stdio.h>
#include <ctype.h>

int main()
{
    char str[200];
    int i, words = 0;
    int inWord = 0;
```

```
    printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);

    for (i = 0; str[i] != '\0'; i++)
    {
        if (isspace((unsigned char)str[i]))
        {
            inWord = 0;
        }
        else if (inWord == 0)
        {
            words++;
            inWord = 1;
        }
    }

    printf("Number of words = %d\n", words);

    return 0;
}
```



14. WRITE A C PROGRAM TO SORT AN ARRAY OF STRING.

```
#include <stdio.h>
#include <string.h>

int main()
{
    char str[50][100];
    char temp[100];
    int n, i, j;
```

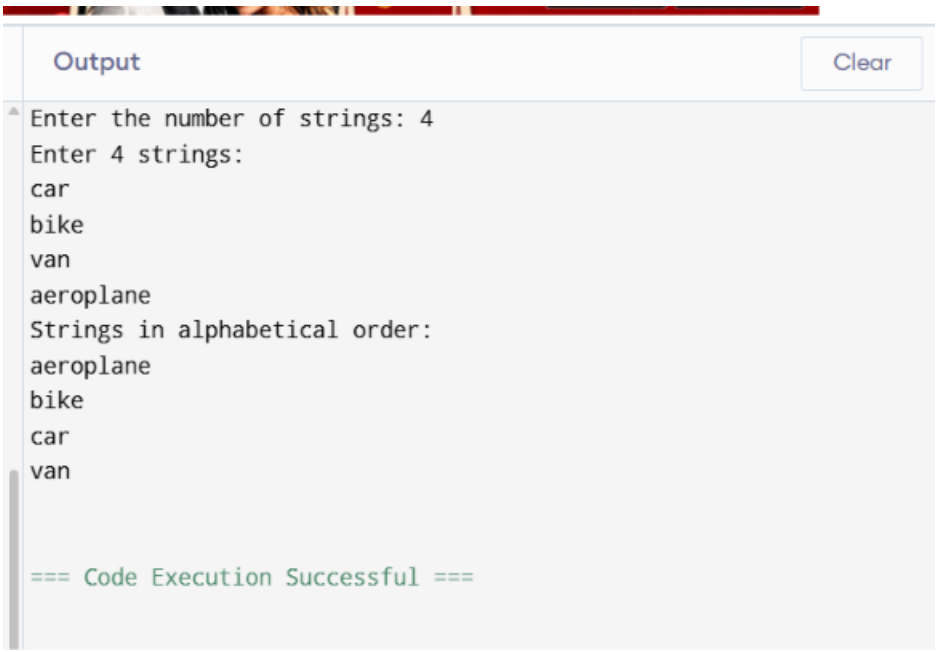
```
    printf("Enter the number of strings: ");
    scanf("%d", &n);

    printf("Enter %d strings:\n", n);
    for (i = 0; i < n; i++)
    {
        scanf("%99s", str[i]);
    }

    for (i = 0; i < n - 1; i++)
    {
        for (j = i + 1; j < n; j++)
        {
            if (strcmp(str[i], str[j]) > 0)
            {
                strcpy(temp, str[i]);
                strcpy(str[i], str[j]);
                strcpy(str[j], temp);
            }
        }
    }

    printf("Strings in alphabetical order:\n");
    for (i = 0; i < n; i++)
    {
        printf("%s\n", str[i]);
    }

    return 0;
}
```



15. WRITE A C PROGRAM TO TOGGLE THE CASE OF SENTENCES(LOWERCASE TO UPPERCASE AND VICEVERSA).

```
#include <stdio.h>

int main()
{
   char str[200];
   int i;

   printf("Enter a sentence: ");
   fgets(str, sizeof(str), stdin);

   for (i = 0; str[i] != '\0'; i++)
   {
      if (str[i] >= 'a' && str[i] <= 'z')
      {
         str[i] = str[i] - 'a' + 'A';
      }
      else if (str[i] >= 'A' && str[i] <= 'Z')
      {
         str[i] = str[i] - 'A' + 'a';
      }
   }

   printf("Sentence after toggling case:\n");
```

```
    printf("%s", str);

    return 0;
}
```

